ASSIGNMENT 4.2

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BATCH-30

Task Description-1

• Zero-shot: Prompt AI with only the instruction. Write a Python function to determine

whether a given number is prime

CODE:

```
def is_prime(n):
    if n < 2:
        return False
    if n == 2:
        return True
    if n % 2 == 0:
        return False
    for i in range(3, int(n**0.5) + 1, 2):
        if n % i == 0:
            return False
    return True

# Get input from user
num = int(input("Enter a number: "))

# Check and display result
if is_prime(num):
    print(f"{num} is a prime number")
else:
    print(f"{num} is not a prime number")
```

OUTPUT:

```
PS D:\AI Assisted coding>  & 'C:\Users\vinee\AppData\Local\Microsoft\WindowsApps\python3.12.exe'
2701-win32-x64\bundled\libs\debugpy\launcher' '64816' '--' 'D:\AI Assisted coding\.vscode\4.5.py'
Enter a number: 15
15 is not a prime number
PS D:\AI Assisted coding>
```

Task Description-2

• One-shot: Provide one example: Input: [1, 2, 3, 4], Output: 10 to help AI generate a function that calculates the sum of elements in a list.

CODE:

```
# Calculate the sum of elements in a list
numbers = [1, 2, 3, 4, 5]
total = sum(numbers)
print(f"Sum of the list: {total}")
```

OUTPUT:

```
PS D:\AI Assisted coding>  & 'C:\Users\vinee\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\U
undled\libs\debugpy\launcher' '52162' '--' 'D:\AI Assisted coding\.vscode\4.5.py'
Sum of the list: 15
```

Task Description-3

• Few-shot: Give 2–3 examples to create a function that extracts digits from an alphanumeric string.

Expected Output-3

• Accurate function that returns only the digits from alphanumeric string.

CODE:

```
def extract_digits(s):
    return ''.join([char for char in s if char.isdigit()])

print(extract_digits("abc123def456"))  # Output: 123456
print(extract_digits("hello42world"))  # Output: 42
```

OUTPUT:

```
PS D:\AI Assisted coding>  d:; cd 'd:\AI Assisted coding'; & 'C:\Users\vinee\AppData\Local\Microsoft\Wi
y-2025.19.2026012701-win32-x64\bundled\libs\debugpy\launcher' '61410' '--' 'D:\AI Assisted coding\.vscod
123456
42
```

Task Description-4

• Compare zero-shot vs few-shot prompting for generating a function that counts the number of vowels in a string.

Expected Output-4

• Output comparison + student explanation on how examples helped the model.

SOLUTION:

Aspect	Zero-Shot Prompting	Few-Shot Prompting
Examples provided	None	One or more
Prompt length	Short	Longer
Output reliability	Good	Very high
Code style control	Limited	Better
Handles edge cases	Depends on model	More consistent
Best used when	Task is simple	Task needs precision
Risk of misinterpretation	Slightly higher	Lower

Task Description-5

• Use few-shot prompting with 3 sample inputs to generate a function that determines the minimum of three numbers without using the built-in min() function.

Expected Output-5

• A function that handles all cases with correct logic based on example patterns.

CODE:

```
def min_of_three(a, b, c):
    """
    Determines the minimum of three numbers without using built-in min().

    Examples:
    >>> min_of_three(5, 2, 8)
    2
    >>> min_of_three(-1, -5, 3)
    -5
    >>> min_of_three(10, 10, 10)
    10
    """
    if a <= b and a <= c:
        return a
    elif b <= a and b <= c:
        return b
    else:
        return c

# Test cases
print(min_of_three(5, 2, 8))      # Output: 2
print(min_of_three(-1, -5, 3))    # Output: -5
print(min_of_three(10, 10, 10))   # Output: 10
```

OUTPUT:

```
PS D:\AI Assisted coding>  d:; cd 'd:\AI Assisted coding'; & 'C:\Users\vinee\AppData\Local\Microsoft\WindowsApps\python3.12.exe' 'c:\Users\vinee\.vscode\ex
y-2025.19.2026012701-win32-x64\bundled\libs\debugpy\launcher' '52399' '--' 'D:\AI Assisted coding\.vscode\4.5.py'
2
-5
10
```